

ALFAAJ19480

## Niacinamide, Thermo Scientific

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

|                               |                                                                                                                                                                                                                                                                    |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 产品说明:<br>Product Description: | Niacinamide, Thermo Scientific<br>Niacinamide, Thermo Scientific                                                                                                                                                                                                   |
| Cat No. :                     | J19480                                                                                                                                                                                                                                                             |
| Synonyms                      | 3-Pyridinecarboxamide; Niacinamide                                                                                                                                                                                                                                 |
| CAS No                        | 98-92-0                                                                                                                                                                                                                                                            |
| Molecular Formula             | C <sub>6</sub> H <sub>6</sub> N <sub>2</sub> O                                                                                                                                                                                                                     |
| Supplier                      | Avocado Research Chemicals Ltd.<br>(Part of Thermo Fisher Scientific)<br>Shore Road, Heysham<br>Lancashire, LA3 2XY,<br>United Kingdom<br>Office Tel: +44 (0) 1524 850506<br>Office Fax: +44 (0) 1524 850608                                                       |
| Emergency Telephone Number    | For information <b>US</b> call: 001-800-227-6701 / <b>Europe</b> call: +32 14 57 52 11<br>Emergency Number <b>US</b> :001-201-796-7100 / <b>Europe</b> : +32 14 57 52 99<br><b>CHEMTREC</b> Tel. No. <b>US</b> :001-800-424-9300 / <b>Europe</b> :001-703-527-3887 |
| E-mail address                | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                                     |
| Recommended Use               | Laboratory chemicals.                                                                                                                                                                                                                                              |
| Uses advised against          | No Information available                                                                                                                                                                                                                                           |

### SECTION 2. HAZARD IDENTIFICATION

|                                                                                                                                           |                     |                              |
|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------------------|
| Physical State<br>Powder Solid                                                                                                            | Appearance<br>White | Odor<br>Practically odorless |
| <b>Emergency Overview</b><br>May be harmful if swallowed. Causes serious eye irritation. May form combustible dust concentrations in air. |                     |                              |

#### Classification of the substance or mixture

|                                   |            |
|-----------------------------------|------------|
| Acute Oral Toxicity               | Category 5 |
| Serious Eye Damage/Eye Irritation | Category 2 |

#### Label Elements



Signal Word

Warning

Hazard Statements

## Niacinamide, Thermo Scientific

H303 - May be harmful if swallowed  
H319 - Causes serious eye irritation

**Precautionary Statements****Prevention**

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P280 - Wear eye protection/ face protection

**Response**

P301 + P312 - IF SWALLOWED: Call a POISON CENTER or doctor if you feel unwell

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P337 + P313 - If eye irritation persists: Get medical advice/attention

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

May form combustible dust concentrations in air.

**Health Hazards**

May be harmful if swallowed. Causes serious eye irritation.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

May form explosible dust-air mixture if dispersed. This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component    | CAS No  | Weight % |
|--------------|---------|----------|
| Nicotinamide | 98-92-0 | >95      |

**SECTION 4. FIRST AID MEASURES****Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention.

**Ingestion**

Do NOT induce vomiting. Get medical attention.

**Most important symptoms and effects**

No information available.

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES**

**Suitable Extinguishing Media**

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Dust can form an explosive mixture with air.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Use personal protective equipment as required. Ensure adequate ventilation.

**Environmental Precautions**

See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Avoid contact with skin and eyes. Do not breathe dust. Avoid ingestion and inhalation. Minimize dust generation and accumulation. Wash hands before breaks and immediately after handling the product.

**Storage**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

## Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Neoprene       | recommendations   |                 |             |                       |
| Natural rubber |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                     |                                 |                                          |
|-------------------------------------|---------------------------------|------------------------------------------|
| <b>Appearance</b>                   | White                           |                                          |
| <b>Physical State</b>               | Powder Solid                    |                                          |
| <b>Odor</b>                         | Practically odorless            |                                          |
| <b>Odor Threshold</b>               | No data available               |                                          |
| <b>pH</b>                           | 6.0-7.5 @ 20°C                  | 5% aq.solution                           |
| <b>Melting Point/Range</b>          | 128 - 131 °C / 262.4 - 267.8 °F |                                          |
| <b>Softening Point</b>              | No data available               |                                          |
| <b>Boiling Point/Range</b>          | 150 - 160 °C / 302 - 320 °F     | @ 760 mmHg                               |
| <b>Flash Point</b>                  | > 150 °C / > 302 °F             | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>             | Not applicable                  | Solid                                    |
| <b>Flammability (solid,gas)</b>     | No information available        |                                          |
| <b>Explosion Limits</b>             | No data available               |                                          |
| <b>Vapor Pressure</b>               | No data available               |                                          |
| <b>Vapor Density</b>                | Not applicable                  | Solid                                    |
| <b>Specific Gravity / Density</b>   | No data available               |                                          |
| <b>Bulk Density</b>                 | No data available               |                                          |
| <b>Water Solubility</b>             | 100g/100ml                      |                                          |
| <b>Solubility in other solvents</b> | No information available        |                                          |

## Partition Coefficient (n-octanol/water)

|                                  |                          |       |
|----------------------------------|--------------------------|-------|
| <b>Component</b>                 | <b>log Pow</b>           |       |
| Nicotinamide                     | -0.38                    |       |
| <b>Autoignition Temperature</b>  | 480 °C / 896 °F          |       |
| <b>Decomposition Temperature</b> | No data available        |       |
| <b>Viscosity</b>                 | Not applicable           | Solid |
| <b>Explosive Properties</b>      | No information available |       |
| <b>Oxidizing Properties</b>      | No information available |       |

|                          |            |
|--------------------------|------------|
| <b>Molecular Formula</b> | C6 H6 N2 O |
| <b>Molecular Weight</b>  | 122.13     |

## SECTION 10. STABILITY AND REACTIVITY

|                                 |                                              |
|---------------------------------|----------------------------------------------|
| <b>Stability</b>                | Stable under recommended storage conditions. |
| <b>Hazardous Reactions</b>      | No information available.                    |
| <b>Hazardous Polymerization</b> | Hazardous polymerization does not occur.     |
| <b>Conditions to Avoid</b>      | Incompatible products. Avoid dust formation. |
| <b>Materials to avoid</b>       | Oxidizing agent.                             |

**Hazardous Decomposition Products** Nitrogen oxides (NOx). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

## (a) acute toxicity;

| Component    | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation |
|--------------|---------------------------|------------------------------|-----------------|
| Nicotinamide | LD50 = 3500 mg/kg ( Rat ) | LD50 > 2000 mg/kg ( Rabbit ) |                 |

(b) skin corrosion/irritation; Based on available data, the classification criteria are not met

(c) serious eye damage/irritation; Category 2

## (d) respiratory or skin sensitization;

|                    |                                                                  |
|--------------------|------------------------------------------------------------------|
| <b>Respiratory</b> | Based on available data, the classification criteria are not met |
| <b>Skin</b>        | Based on available data, the classification criteria are not met |

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

(f) carcinogenicity; Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; Based on available data, the classification criteria are not met

(h) STOT-single exposure; Based on available data, the classification criteria are not met

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

## Niacinamide, Thermo Scientific

**Target Organs** None known.

**(j) aspiration hazard;** Not applicable  
Solid

**Symptoms / effects, both acute and delayed** No information available

## SECTION 12. ECOLOGICAL INFORMATION

## Ecotoxicity effects

| Component    | Freshwater Fish                                     | Water Flea | Freshwater Algae | Microtox |
|--------------|-----------------------------------------------------|------------|------------------|----------|
| Nicotinamide | LC50: > 1000 mg/L, 96h static (Poecilia reticulata) |            |                  |          |

**Persistence and Degradability** Readily biodegradable  
**Persistence** Soluble in water, Persistence is unlikely, based on information available.

**Bioaccumulative Potential** Bioaccumulation is unlikely

| Component    | log Pow | Bioconcentration factor (BCF) |
|--------------|---------|-------------------------------|
| Nicotinamide | -0.38   | No data available             |

**Mobility in soil** The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors.  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance.  
**Ozone Depletion Potential** This product does not contain any known or suspected substance.

## SECTION 13. DISPOSAL CONSIDERATIONS

**Waste from Residues/Unused Products** Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

**Other Information** Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

## SECTION 14. TRANSPORT INFORMATION

Road and Rail Transport

IMDG/IMO Not regulated

IATA Not regulated

## Niacinamide, Thermo Scientific

Special Precautions for User No special precautions required

## SECTION 15. REGULATORY INFORMATION

## International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component    | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|--------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Nicotinamide | -                                                   | -                                       | X    | X     | 202-713-4 | X    | X   | X     | X    | X    | X    | KE-29935 |

## National Regulations

## SECTION 16. OTHER INFORMATION

Prepared By Health, Safety and Environmental Department  
 Creation Date 26-Sep-2009  
 Revision Date 29-Apr-2024  
 Revision Summary New emergency telephone response service provider.

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

PNEC - Predicted No Effect Concentration

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (Volatile Organic Compound)

## Key literature references and sources for data

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<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

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**End of Safety Data Sheet**